Linear Models for Regression

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Outline

- Ordinary least squares
- Model selection and shrinkage
  - Ridge regression
  - LASSO
  - Etc.

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Ordinary Least Squares

Formulation

- Observe a collection of i.i.d. **training data**

$$(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2), \ldots, (\boldsymbol{x}_n, y_n) \text{ from } \mathcal{D}.$$

- **Predictors** (covariates, inputs, features, regressors):

$$\boldsymbol{x}_i = (x_{i1}, \ldots, x_{ip}).$$

- **Response** (outcome, output, regressand):

$$y_i \in \mathbb{R}.$$

真不可控+我不想控制的

- Assume $Y = f(\boldsymbol{X}) + \varepsilon$. We want to build a function $\hat{f}$ to approximate f, so that when given a new input vector $\boldsymbol{x}$, we can predict the output $\hat{y} = \hat{f}(\boldsymbol{x})$.

Formulation

- **Loss**: measures the discrepancy of Y and $\hat{f}(\boldsymbol{X})$.
- Squared error loss: $L\left(Y, \hat{f}(\boldsymbol{X})\right) = \left(Y - \hat{f}(\boldsymbol{X})\right)^2$.
- **Risk**: expected loss

$$R(\hat{f}) = E_D\left(Y - \hat{f}(\boldsymbol{X})\right)^2.$$

由x和e同时导致的loss期望

- Optimal prediction:

$$\begin{aligned} f^* &= \operatorname{argmin}_{\hat{f}} R(\hat{f}) \\ &= \operatorname{argmin}_{\hat{f}} E_D\left(Y - \hat{f}(\boldsymbol{X})\right)^2 \\ &= E(Y|X) = f \end{aligned}$$

Empirical Risk Minimization

- Since D is unknown, we cannot evaluate $R(\hat{f})$.
- However, the training data $(\boldsymbol{x}_1, y_1), \ldots, (\boldsymbol{x}_n, y_n)$ are i.i.d. from D.
- Replace the expectation over D by the average over the training samples, i.e., minimize the training error:

$$R_{emp}(\hat{f}) = \frac{1}{n}\sum_{i=1}^{n} L\left(y_i, \hat{f}(\boldsymbol{x}_i)\right).$$

- Least squares estimation

$$\min_{\hat{f}} \sum_{i=1}^{n} \left(y_i - \hat{f}(\boldsymbol{x}_i)\right)^2.$$

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Linear Regression

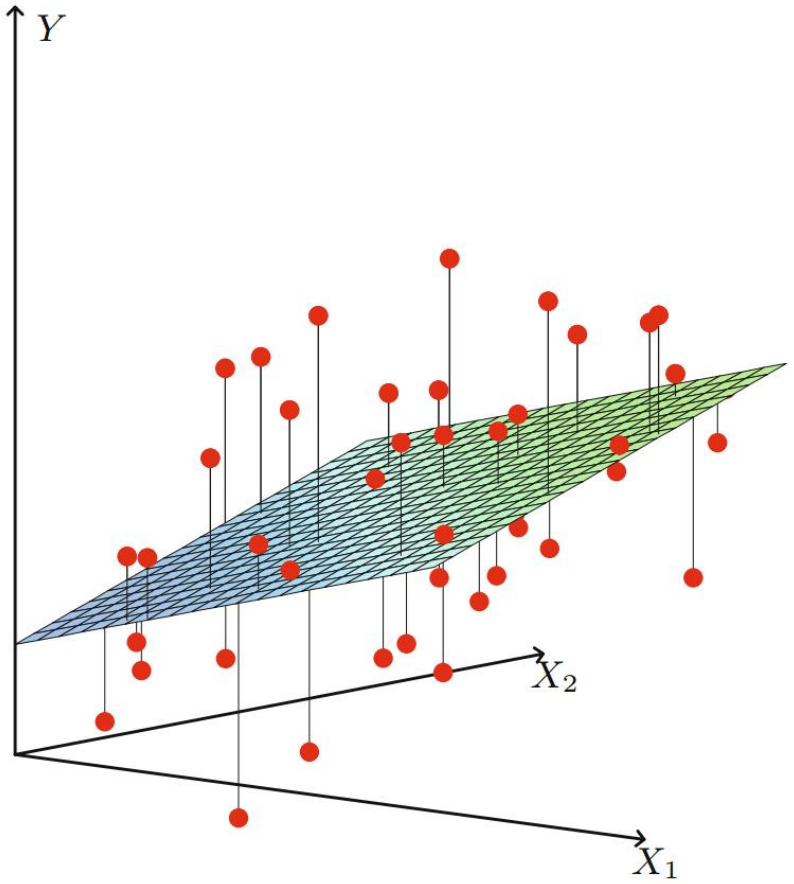
- f: the underlying truth. *Unknown*.
- We have to put some restrictions, i.e., assume some structures on f.
- Assumption

$$f(\boldsymbol{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p,$$

where β_j's are **unknown parameters** and β_0 is the **intercept**.

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Linear Regression



Linear Regression

- What if the model is wrong?
  - George Box: all models are wrong, but some are useful.
  - John von Neumann: truth … is much too complicated to allow anything but approximations.

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Least Squares Regression

- Estimation of f is reduced to estimation of $\boldsymbol{\beta}$.
- Ordinary least squares estimation (OLS):

$$\underset{\boldsymbol{\beta}}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - f(\boldsymbol{x}_i)\right)^2$$

$$= \underset{\boldsymbol{\beta}}{\operatorname{argmin}} \, (\boldsymbol{y} - X\boldsymbol{\beta})^T (\boldsymbol{y} - X\boldsymbol{\beta})$$

where $\boldsymbol{y}$ is an n-dimensional vector of response values, X is an $n \times (p+1)$ matrix of predictors, and $\boldsymbol{\beta}$ is a $(p+1)$-dimensional vector of parameters.

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Estimate $\boldsymbol{\beta}$

- Differentiate the criterion with respect to $\boldsymbol{\beta}$ and set the derivative equal to 0, we get the solution:

$$\widehat{\boldsymbol{\beta}} = \left(X^T X\right)^{-1} X^T \boldsymbol{y}$$

  - The “hat” notation: used for estimates.
  - X full rank $\Leftrightarrow$ $X^T X$ invertible.

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Computing $\hat{\beta}$


```
b.hat = (t(X) %*% X)^-1 %*% t(X) %*% y
b.hat = solve(t(X) %*% X) %*% t(X) %*% y
b.hat = solve(t(X) %*% X) %*% (t(X) %*% y)
b.hat = solve(t(X) %*% X, t(X) %*% y)
```

- Are all of these correct?
- Which is/are the best?

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The Hat Matrix

- $\hat{y} = X\hat{\beta} = X\left(X^T X\right)^{-1} X^T y = Hy$, where

$$H = X\left(X^T X\right)^{-1} X^T$$

  is called the Hat-matrix.

- H is an orthogonal projection matrix.

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Projection Matrix

- Definition:
  - A projection matrix P is a square matrix that is idempotent, i.e.,

$$P^2 = P.$$

  - An orthogonal projection matrix P is one that satisfies $range(P) \perp null(P)$. (Note: orthogonal projection matrices are not orthogonal matrices!)
  - Theorem: A projection P is an orthogonal projection if and only if $P = P^T$.

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Projection Matrix

- If $u \in \mathbb{R}^n$and $\|u\| = 1$, then $P = uu^T$ is the orthogonal projection to the 1-dimensional subspace spanned by u.

- If U has orthonormal columns, then $P = UU^T$is the orthogonal projection to the column space of U.

- For an arbitrary $n \times p$ matrix X (with full rank), $P = X\left(X^TX\right)^{-1}X^T$ is the orthogonal projection to the column space of X.

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The Projection Interpretation of OLS

- It is easy to see that

$$H = X\left(X^T X\right)^{-1} X^T$$

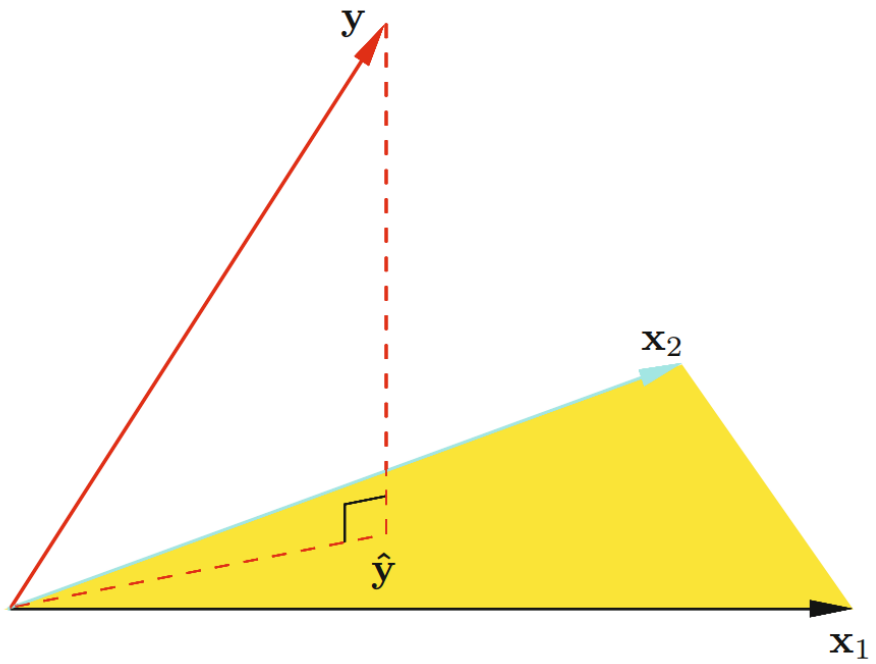
  is an orthogonal projection matrix.

- $\hat{y} = Hy$ is y projected to the column space of X.

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The Projection Interpretation of OLS

$$\hat{y} = X\left(X^T X\right)^{-1} X^T y$$



- The outcome vector y is orthogonally projected onto the subspace spanned by the input vectors x_1 and x_2.

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Properties of $\hat{\beta}$

- We often assume further that the errors ε_i are i.i.d. with $E(\varepsilon_i) = 0$ and $\mathrm{Var}(\varepsilon_i) = \sigma^2$.
- Unbiased: $E(\hat{\beta}) = \beta$.
- Variance-covariance

$$Cov(\hat{\beta}) = (X^T X)^{-1}\sigma^2.$$

- Typically,

$$\hat{\sigma}^2 = \frac{1}{N-p-1}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2.$$

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Sum of Squares

- $SS_{tot} = \sum_{i=1}^{n}(y_i - \bar{y})^2$
- $SS_{reg} = \sum_{i=1}^{n}(\hat{y}_i - \bar{y})^2$
- $SS_{res} = \sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = SS_{tot} - SS_{reg}$
- $R^2 = 1 - \frac{SS_{res}}{SS_{tot}} = \frac{SS_{reg}}{SS_{tot}}$ 由回归解释的变差 / 总变差

Properties of $\hat{\beta}$

- Let R_j^2 be the R^2 from the linear regression

$$X_j = \alpha_0 + \sum_{k \neq j} \alpha_k X_k + \varepsilon.$$

- Let $S_{x_j}^2 = \frac{1}{n-1} \sum_{i=1}^{n} \left(x_{ij} - \overline{x_j} \right)^2$.
- Then

$$\mathrm{Var}\left(\hat{\beta}_j\right) = \frac{\sigma^2}{(n-1) S_{x_j}^2} \cdot \frac{1}{1-R_j^2}.$$

- The part $\frac{1}{1-R_j^2}$ is called the **variance inflation factor**.

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The Gauss-Markov Theorem

- Assumption: $y = X\beta + \varepsilon$, where $E(\varepsilon) = 0, Cov(\varepsilon) = \sigma^2 I$.
- Theorem: If $\hat{\beta}$ is the OLS estimate of β, and if β^* is any other linear unbiased estimate of β (i.e., $\beta^* = By$ and $E(\beta^*) = \beta$), then

$$\mathrm{Var}\left(c^T\hat{\beta}\right) \leq \mathrm{Var}\left(c^T\beta^*\right)$$

for any $c \in \mathbb{R}^{p+1}$.

  - Best Linear Unbiased Estimator (BLUE).
  - Example: Make a prediction at $x^* = \left(x_1^*, \ldots, x_p^*\right)^T$. Let $c = \left(1, x_1^*, \ldots, x_p^*\right)^T$, then $E(Y|x^*) = c^T\beta$.

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The Gauss-Markov Theorem

- If the linear model is correct for a given problem, then the least squares prediction is unbiased, and has the lowest variance among all unbiased estimators that are linear functions of y.

- But there can be biased estimators with smaller variance.

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Prediction Accuracy

- Consider $\hat{f}(x^*)$ as an estimate of $E(Y|x^*) = f(x^*)$, then the Mean Squared Error (MSE)

$$MSE\left(\hat{f}(x^*)\right) = E\left(f(x^*) - \hat{f}(x^*)\right)^2$$

$$= \left(f(x^*) - E\left(\hat{f}(x^*)\right)\right)^2 + Var\left(\hat{f}(x^*)\right)$$

$$= Bias^2 + Variance.$$

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Prediction Accuracy

- Choosing estimators often involves a trade-off between bias and variance.

- Generally, by *regularizing* (shrinking, dampening, controlling) the estimator in some way, its variance will be reduced; if the corresponding increase in bias is small, this will be worthwhile.

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Example: Prostate Cancer


```
library(ElemStatLearn)
str(prostate)
```

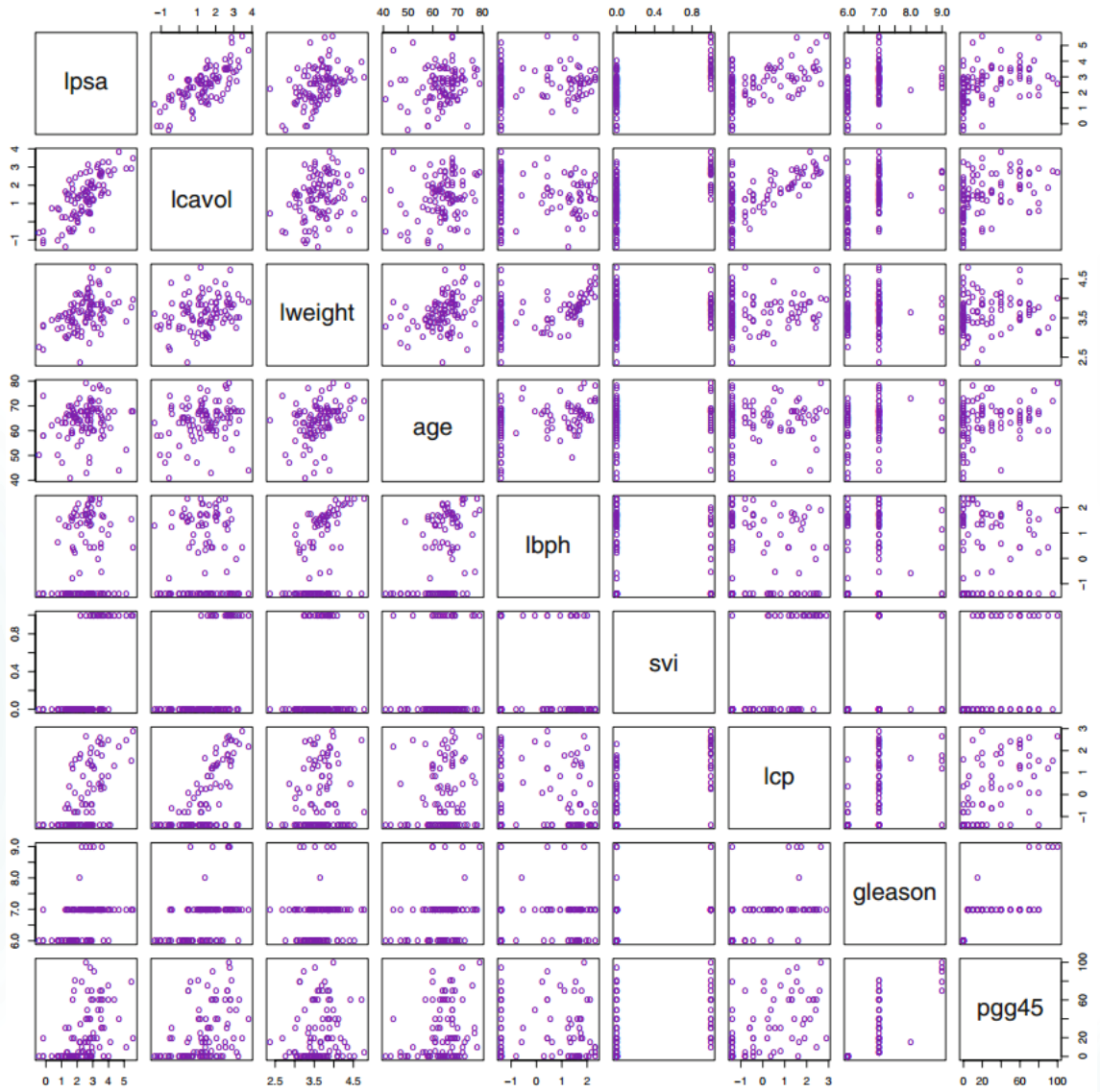
```
## 'data.frame': 97 obs. of 10 variables:
## $ lcavol : num -0.58 -0.994 -0.511 -1.204 0.751 ...
## $ lweight: num 2.77 3.32 2.69 3.28 3.43 ...
## $ age : int 50 58 74 58 62 50 64 58 47 63 ...
## $ lbph : num -1.39 -1.39 -1.39 -1.39 -1.39 ...
## $ svi : int 0 0 0 0 0 0 0 0 0 0 ...
## $ lcp : num -1.39 -1.39 -1.39 -1.39 -1.39 ...
## $ gleason: int 6 6 7 6 6 6 6 6 6 6 ...
## $ pgg45 : int 0 0 20 0 0 0 0 0 0 0 ...
## $ lpsa : num -0.431 -0.163 -0.163 -0.163 0.372 ...
## $ train : logi TRUE TRUE TRUE TRUE TRUE TRUE ...
```

Example: Prostate Cancer

- objects: men who were about to receive a radical prostatectomy
  - lpsa: level of prostate specific antigen (ng/mL, normal range 0-2.1)
  - lcavol: logarithm of cancer volume
  - lweight: logarithm of prostate weight
  - age
  - lbph: logarithm of the amount of benign prostatic hyperplasia
  - svi: seminal vesicle invasion (means that the tumor has spread to the seminal vesicles)
  - lcp: logarithm of capsular penetration
  - gleason: Gleason score (Gleason scores range from 2 to 10 and indicate how likely it is that a tumor will spread.)
  - pgg45: percent of Gleason scores 4 or 5

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Example: Prostate Cancer



Example: Prostate Cancer

	lcavol	lweight	age	lbph	svi	lcp	gleason
lweight	0.300						
age	0.286	0.317					
lbph	0.063	0.437	0.287				
svi	0.593	0.181	0.129	−0.139			
lcp	0.692	0.157	0.173	−0.089	0.671		
gleason	0.426	0.024	0.366	0.033	0.307	0.476	
pgg45	0.483	0.074	0.276	−0.030	0.481	0.663	0.757

Correlation coefficients of the predictors

Ridge Regression

Ridge Regression

- When there are many highly correlated variables

需要约束线性相关性较大时的系数方差放大

  - $\hat{\beta}_{ols}$ may have a large coefficient on one variable and a similarly large negative coefficient on its correlated variable (Unstable).

  - In ridge regression, the size constraint tries to avoid this phenomenon.

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Ridge Regression

岭回归

- Restrict the squared sum (l^2 norm) of the coefficients

$$\min_{\beta} \ \sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2$$

$$\text{s.t.} \ \sum_{j=1}^{p}\beta_j^2 \le s$$

常数项的方差放大不关注

  - $s \ge 0$ is a *tuning parameter* that controls the level of regularization.

约束

  - What happens if $s \to 0$?
  - What happens if $s \to \infty$?

Equivalent Formulation

$$\hat{\beta}^{ridge} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 .$$

- Constrained optimization becomes unconstrained.
- Why are the formulations equivalent?

Karush-Kuhn-Tucker Necessary Conditions

- Theorem: Let X be a nonempty open set in $\mathbb{R}^n$, and let $f: \mathbb{R}^n \to \mathbb{R}$ and $g_i: \mathbb{R}^n \to \mathbb{R}$ for $i = 1, \ldots, m$. Consider

Problem P

$$\min_x \ f(x)$$
$$\text{s.t.} \ \ g_i(x) \leq 0 \quad \text{for } i = 1, \ldots, m,$$
$$x \in X.$$

不等式约束记为g
等式约束记为h

Karush-Kuhn-Tucker Necessary Conditions

可行解

- Let $\bar{x}$ be a feasible solution, and denote $I = \{i: g_i(\bar{x}) = 0\}$. Suppose that f and $\{g_i\}_{i=1}^m$ are differentiable at $\bar{x}$, and that $\nabla g_i(\bar{x})$ for i $\in I$ are linearly independent. If $\bar{x}$ solves Problem **P** locally, there exist unique scalars $\{u_i\}$ such that

$$\nabla f(\bar{x}) + \sum_{i=1}^m u_i \nabla g_i(\bar{x}) = 0$$

$$u_i g_i(\bar{x}) = 0 \quad \text{for } i = 1, \dots m$$

互补松弛条件

$$u_i \geq 0 \quad \text{for } i = 1, \dots m$$

Karush-Kuhn-Tucker Necessary Conditions

- If Problem **P** further has equality constraints:

$$\begin{aligned} \min_{x} \quad & f(x) \\ \text{s.t.} \quad & g_i(x) \leq 0 \quad \text{for } i = 1, \ldots, m, \\ & h_i(x) = 0 \quad \text{for } i = 1, \ldots, l, \\ & x \in X. \end{aligned}$$

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Karush-Kuhn-Tucker Necessary Conditions

- For a feasible solution $\bar{x}$, further assume that each h_i for $i = 1, \ldots, l$ is continuously differentiable at $\bar{x}$, and that $\nabla g_i(\bar{x})$ for i $\in I$ and $\nabla h_i(\bar{x})$ for $i = 1, \ldots, l$ are linearly independent. Then if $\bar{x}$ solves Problem **P** locally, there exist unique scalars u_i and v_i such that

$$\nabla f(\bar{x}) + \sum_{i=1}^{m} u_i \nabla g_i(\bar{x}) + \sum_{i=1}^{l} v_i \nabla h_i(\bar{x}) = 0$$

$$u_i g_i(\bar{x}) = 0 \quad \text{for } i = 1, \ldots m$$

$$u_i \geq 0 \quad \text{for } i = 1, \ldots m$$

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Sufficient Conditions

- When is a local minimum globally optimal?

- Convexity will be sufficient, which includes
  - convexity of the target function
  - convexity of the feasible region

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Karush-Kuhn-Tucker Sufficient Conditions (Simplified Ver.)

- Let $\bar{x}$ be a feasible solution, and suppose $\bar{x}$ together with multipliers (u, v) satisfies

$$\nabla f(\bar{x}) + \nabla g(\bar{x})^T u + \nabla h(\bar{x})^T v = 0,$$

$$u \geq 0,$$

$$u_i g_i(\bar{x}) = 0, \quad i = 1, \dots m.$$

- If f is a pseudoconvex function, g_i, $i = 1, \dots, m$ are quasiconvex functions, and h_i, $i = 1, \dots, l$ are affine functions, then $\bar{x}$ is a global optimal solution.

Solution for Ridge Regression

岭回归

$$\hat{\beta}^{ridge} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 .$$

- The ridge solutions are not equivariant under scaling of the inputs.
- Thus, it makes sense to center and standardize the inputs first.
- $\hat{\beta}_0 = \bar{y}; y = y - \bar{y}.$
- $\hat{\beta}^{ridge} = \underset{\beta}{\operatorname{argmin}} (y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta.$

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Solution for Ridge Regression

- Ridge solution:

$$\hat{\beta}^{ridge} = \left(X^T X + \lambda I\right)^{-1} X^T y.$$

- $\hat{\beta}^{ridge}$ is linear in y.
- $\hat{\beta}^{ridge}$ is biased.
- Even if X is not full-rank, $\left(X^T X + \lambda I\right)$ is still invertible.

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Shrinkage in Ridge

- Let X has singular value decomposition

$$X = UDV^T,$$

where U and V are $N \times p$ and $p \times p$ matrices with orthonormal columns, and $D = \text{diag}\{d_1, \ldots, d_p\}$ with $d_1 \geq d_2 \geq \cdots \geq d_p \geq 0$.

- U spans the column space of X (orthonormal basis of $\text{col}(X)$).
- V spans the row space of X (orthonormal basis of $\text{row}(X)$).

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Shrinkage in Ridge

- Then $X^T X$ has (eigen/spectral) decomposition

$$X^T X = V \begin{pmatrix} d_1^2 & & \\ & \ddots & \\ & & d_p^2 \end{pmatrix} V^T.$$

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Shrinkage in Ridge

- Then

$$\hat{\beta}^{ridge} = \left(X^T X + \lambda I\right)^{-1} X^T y$$

$$= \left(X^T X + \lambda I\right)^{-1} X^T X \cdot \hat{\beta}^{ols}$$

$$= V \begin{pmatrix} \frac{d_1^2}{d_1^2+\lambda} & & \\ & \ddots & \\ & & \frac{d_p^2}{d_p^2+\lambda} \end{pmatrix} V^T \cdot \hat{\beta}^{ols}$$

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Shrinkage in Ridge

- $\hat{\beta}^{ridge}$ has smaller variance than $\hat{\beta}^{ols}$, thus may have smaller MSE.

$$Cov(\hat{\beta}^{ridge}) = \sigma^2(X^TX + \lambda I)^{-1}X^TX(X^TX + \lambda I)^{-1}$$

$$= \sigma^2 V \begin{pmatrix} \frac{d_1^2}{(d_1^2+\lambda)^2} & & \\ & \ddots & \\ & & \frac{d_p^2}{(d_p^2+\lambda)^2} \end{pmatrix} V^T.$$

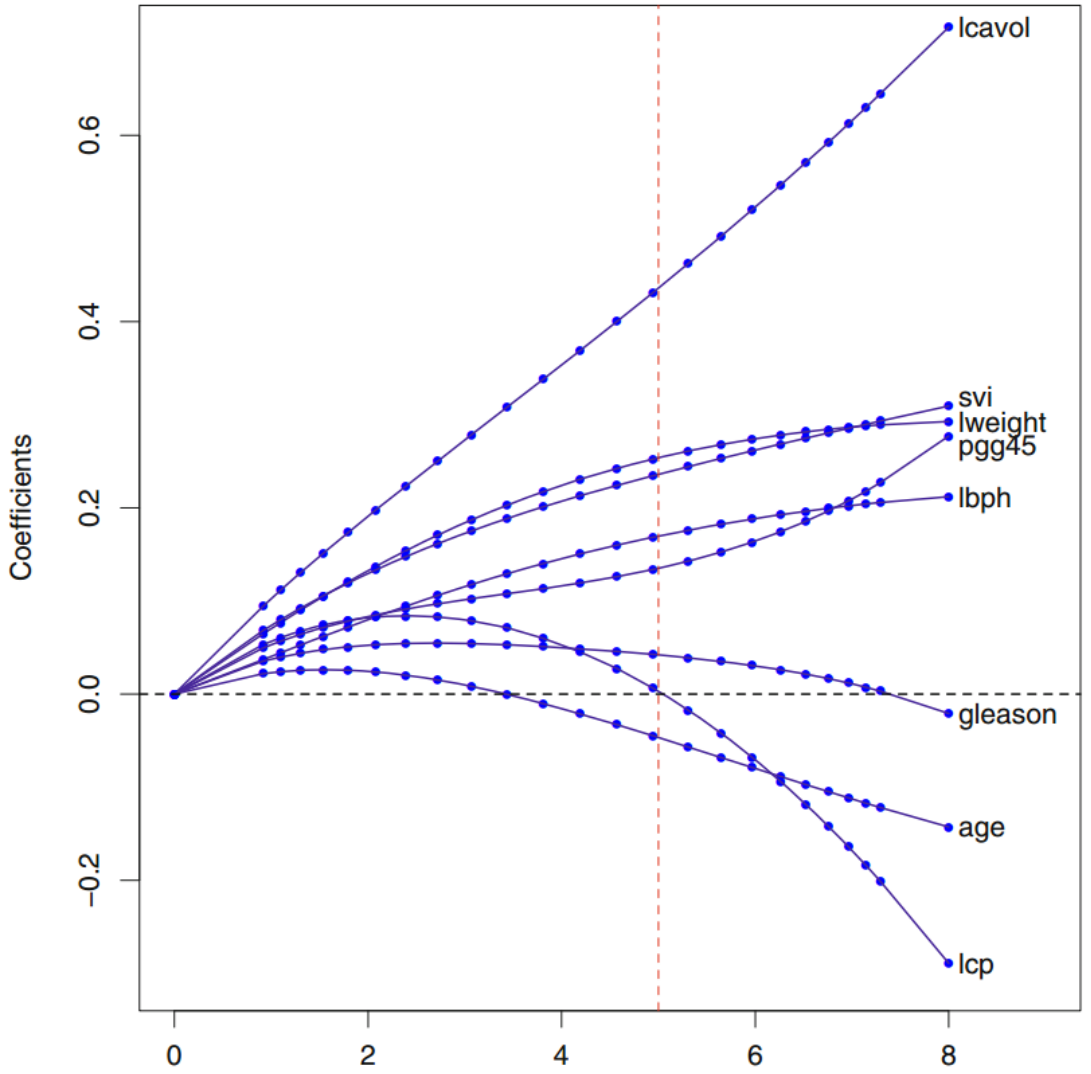
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Shrinkage in Ridge

- If $X^T X = I$ (orthogonal design), then $\hat{\beta}^{ridge} = \frac{1}{1+\lambda}\hat{\beta}^{ols}$.
- Shrink the estimate towards zero by a positive constant less than 1.
- $Var(\hat{\beta}_j^{ridge}) = \frac{1}{(1+\lambda)^2} Var(\hat{\beta}_j^{ols})$.
- $\lambda \uparrow$, shrinkage $\uparrow$, bias $\uparrow$, variance $\downarrow$;
- $\lambda \downarrow$, shrinkage $\downarrow$, bias $\downarrow$, variance $\uparrow$.

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Ridge Solution Path



有效自由度，degree of freedom

Model Assessment

- Objectives:
  - 1. Choose a value of a tuning parameter.
  - 2. Estimate the prediction performance of a given model.

- For both of these purposes, the best approach is to run the procedure on an independent test data, if one is available.

- If possible one should use different test data for (1) and (2) above: a validation set for (1) and a test set for (2).

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Cross-Validation

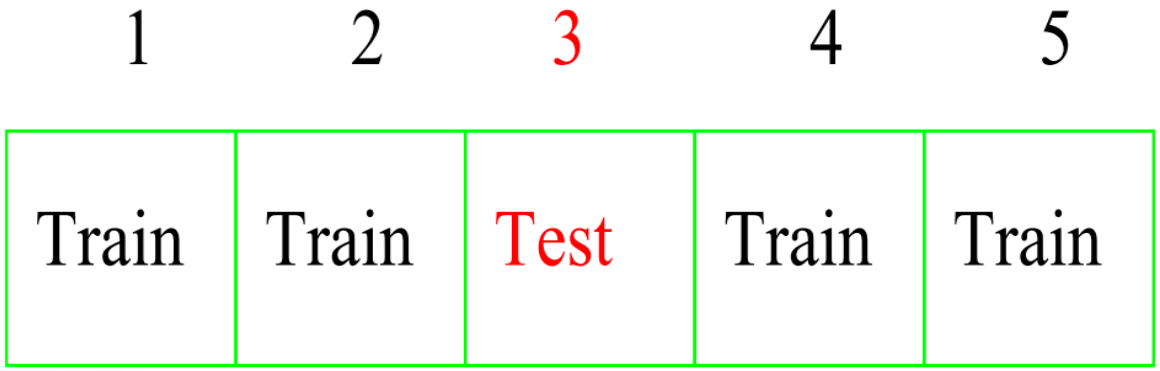
交叉验证的目的是评估模型的水平，而不是为了得到最终的bata

- Often there is insufficient data to create a separate validation or test set; setting some data aside for validation is possible, but affects the accuracy of training estimates.

- In this case, *cross-validation* is useful.

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K-Fold Cross-Validation



- 1. Divide the data into K disjoint subsets.
- 2. Use subsets 2,…, K as training data and subset 1 as validation data. Compute the prediction error on subset 1.
- 3. Repeat for each subset.
- 4. Average the result.

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K-Fold Cross-Validation


```
group.id = rep(1:K, length.out=N)
group.id = sample(group.id,N)
Err = rep(0,K)
for (k in 1:K) {
       Fit model with dat[group.id!=k,]
       Err[k] = Error on dat[group.id==k,]
}
Err.kfold = mean(Err)
```

K-Fold Cross-Validation

- Bias-variance tradeoff in estimating error
  - *K* large: lower bias (large training sets), higher variance.
  - *K* small: higher bias (small training sets), lower variance.
  - Typically choose $K = 5$ or 10.

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Question

- Suppose you use 10-fold cross-validation, you obtain 10 estimates of β, which one should you keep as the final $\hat{\beta}$?

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自助法

0.632 Bootstrap Cross-Validation

评估方法的稳健性

- Bootstrap: sample N times from N samples with replacement.
- Bootstrapped samples contain duplications. On average, a bootstrapped sample contains 63.2% of the original sample.
- Suppose we take B bootstrap samples from the training data. Fit model $\hat{f}^b$ for $b = 1, \ldots, B$.

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0.632 Bootstrap Cross-Validation

- The leave-one-out bootstrap estimator mimics cross-validation and is defined as:

$$Err_{(1)} = \frac{1}{N}\sum_{i=1}^{N}\frac{1}{\left|C^{-i}\right|}\sum_{b\in C^{-i}} L\left(y_i, \hat{f}^b(x_i)\right),$$

where C^{-i} is the set of indices for the bootstrap samples that do not contain observation i, and $\left|C^{-i}\right|$ is the number of such samples.

- Efron and Tibshirani proposed the .632 bootstrap estimator:

$$Err_{0.632} = 0.368 \times Err_{train} + 0.632 \times Err_{(1)}.$$

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0.632 Bootstrap Cross-Validation


```
Fit model with dat
Err.train = training error
Err.boot = matrix(NA, nrow=N, ncol=B)
for (b in 1:B) {
      id.t = sample(1:N, N, replace=T)
      id.v = set.diff(1:N, unique(id.t))
      Fit model with dat[id.t,]
      Err.boot[id.v,b] = error on dat[id.v,]
}
Err.632 = mean(rowMeans(Err.boot, na.rm=T))*0.632 + Err.train*0.368
```

Question

- How to select value for the tuning parameter using cross-validation?

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Cross-Validation: Right and Wrong

- Consider fitting a model on a training data:
  - 1. Starting with 5000 predictors, find the 100 predictors having the largest correlation with the outcome.
  - 2. Fit a ridge regression using the 100 selected predictors.
- How do we estimate the performance of this method with cross-validation?
- Can we apply cross-validation in step 2, forgetting about step 1?

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Cross-validation in deep learning

- Cross-validation is not always used in deep learning. Why?

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LASSO

Variable Selection

- Parsimony: we would like to determine a smaller subset of predictors that exhibit the strongest effects.
  - Best subset selection
  - Forward selection
  - Backward elimination
  - Stepwise selection
- May have lower prediction error than the full model.
- Tends to be myopic at each step.
- The selection process is discrete -- unstable.

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LASSO

- Least absolute shrinkage and selection operator (Chen, Donoho and Saunders 1996; Tibshirani 1996)

$$\hat{\beta}^{lasso} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| .$$

  - Shrinkage
  - Sparsity: some fitted coefficients are exactly zero
  - Continuous variable selection

Equivalent Formulation

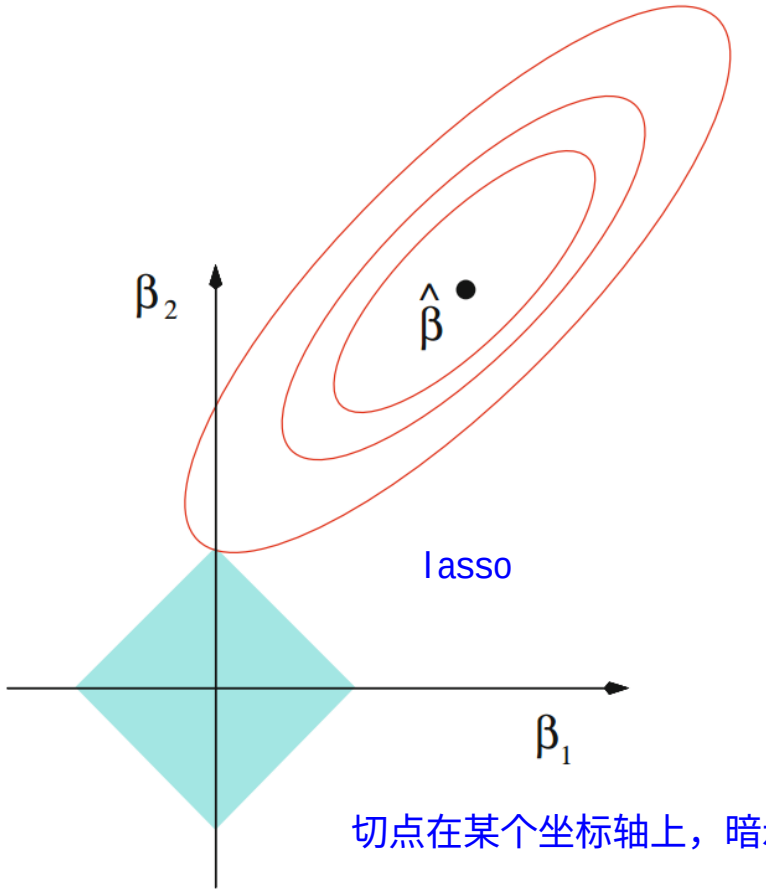
$$\hat{\beta}^{lasso} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

$$\text{s.t. } \sum_{j=1}^{p} |\beta_j| \leq s$$

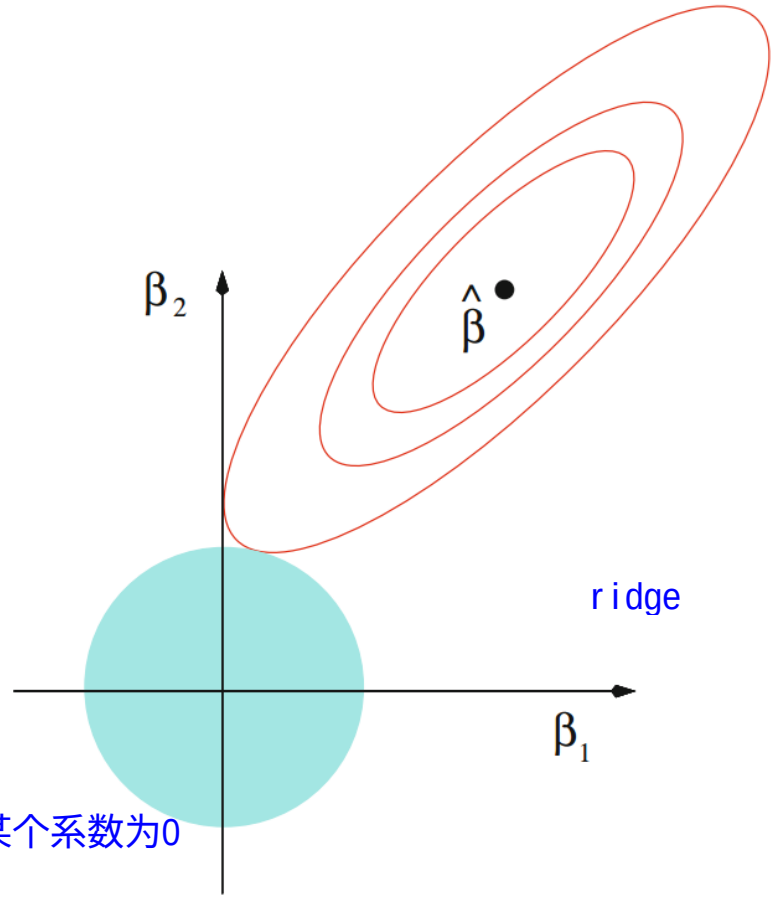
- Can be shown with the Duality Theorem (differentiability not required).

Sparsity

解的稀疏



lasso



ridge

切点在某个坐标轴上，暗示某个系数为0

Soft Thresholding

- When X is orthonormal, observe that

$$(y - X\beta)^T(y - X\beta) = \text{constant} + \sum_{j=1}^{p} \left(\beta_j - \hat{\beta}_j^{ols}\right)^2$$

- Thus, we can minimize over β componentwise

$$\hat{\beta}_j^{lasso} = \underset{\beta_j}{\text{argmin}} \left(\beta_j - \hat{\beta}_j^{ols}\right)^2 + \lambda|\beta_j|.$$

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Soft Thresholding

- The solution is

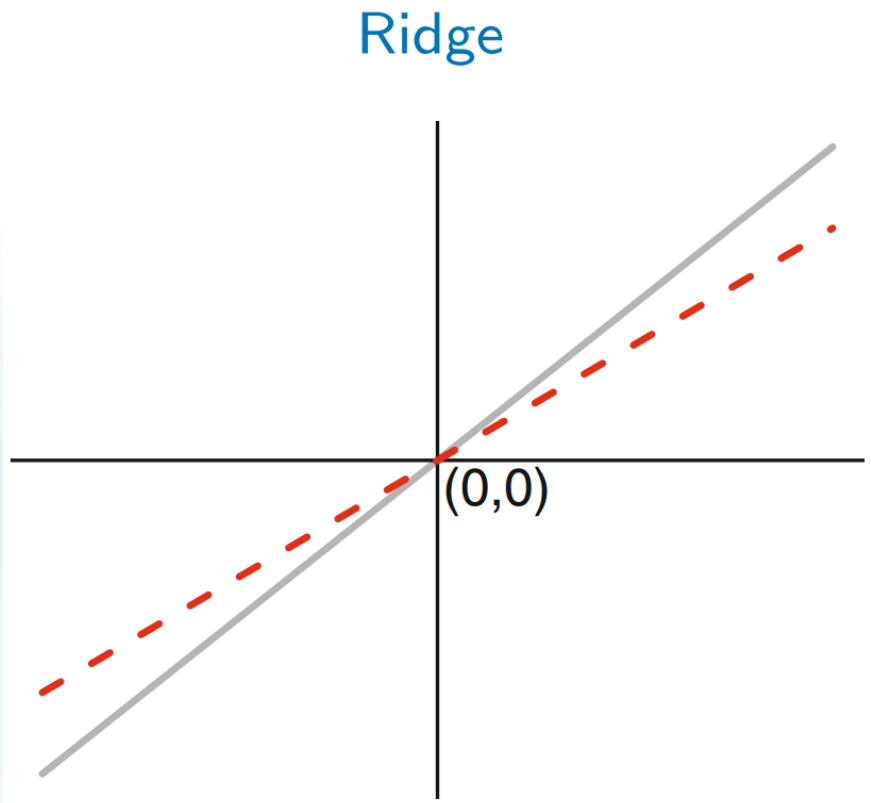
$$\hat{\beta}_j^{lasso} = \begin{cases} \hat{\beta}_j^{ols} - \frac{\lambda}{2}, & if\ \ \hat{\beta}_j^{ols} > \frac{\lambda}{2} \\ 0, & if\ \ |\hat{\beta}_j^{ols}| \leq \frac{\lambda}{2} \\ \hat{\beta}_j^{ols} + \frac{\lambda}{2}, & if\ \ \hat{\beta}_j^{ols} < -\frac{\lambda}{2} \end{cases}$$

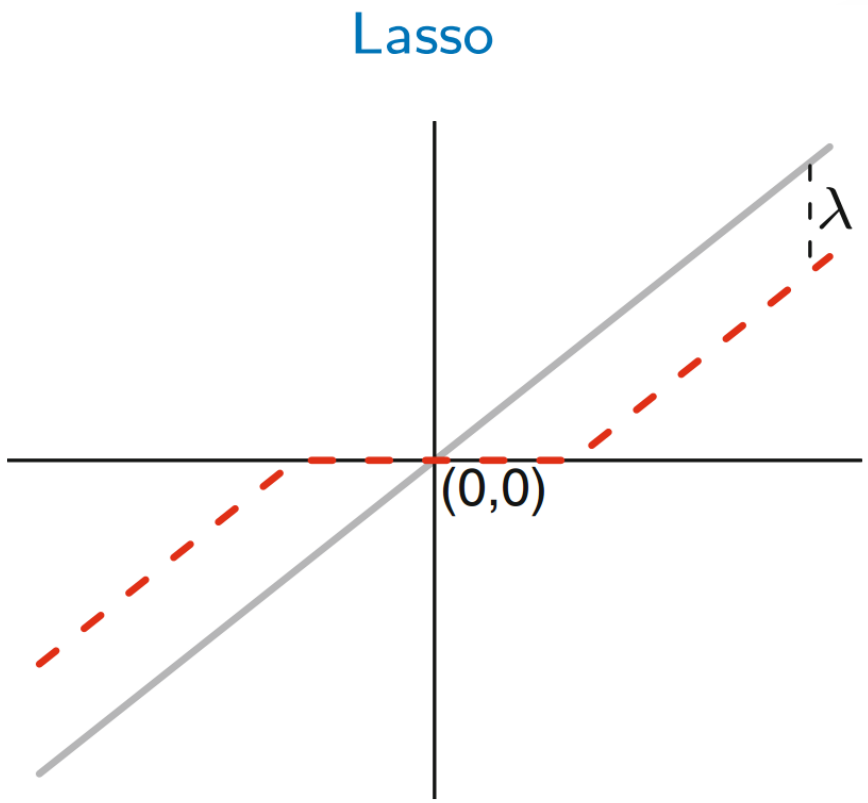
$$= \text{sign}\left(\hat{\beta}_j^{ols}\right) \cdot \left(\left|\hat{\beta}_j^{ols}\right| - \frac{\lambda}{2}\right)_+$$

  - Lasso shrinks large coefficients by a constant.
  - Lasso truncates small coefficients to zero.

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Lasso vs. Ridge





Computation

- Convex programming: write β_j as $\beta_j^+ - \beta_j^-$. $\beta_j^+, \beta_j^- \geq 0$.

$$\min_{\beta_0, \left\{\beta_j^{+/-}\right\}_{j=1}^{p}} \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \left(\beta_j^+ - \beta_j^-\right) x_{ij}\right)^2 + \lambda \left(\beta_j^+ + \beta_j^-\right)$$

$$\text{subject to} \quad \left\{\beta_j^+\right\}_{j=1}^{p}, \left\{\beta_j^-\right\}_{j=1}^{p} \geq 0$$

Computation

- LARS (Efron et al. 2004): The solution path $\{\hat{\beta}(\lambda), 0 \leq \lambda < \infty\}$ is piecewise linear in λ.
  1. Standardize the predictors to have mean zero and unit norm. Start with the residual $r = y - \bar{y}, \beta_1, \ldots, \beta_p = 0$.
  2. Find the predictor x_j most correlated with r.
  3. Move β_j from 0 towards its least-squares coefficient $\langle x_j, r \rangle$, until some other competitor x_k has as much correlation with the current residual as does x_j.
  4. Move β_j and β_k in the direction defined by their joint least squares coefficient of the current residual on (x_j, x_k), until some other competitor x_l has as much correlation with the current residual.
  5. Continue in this way until all p predictors have been entered. After $\min(N -$

Computation

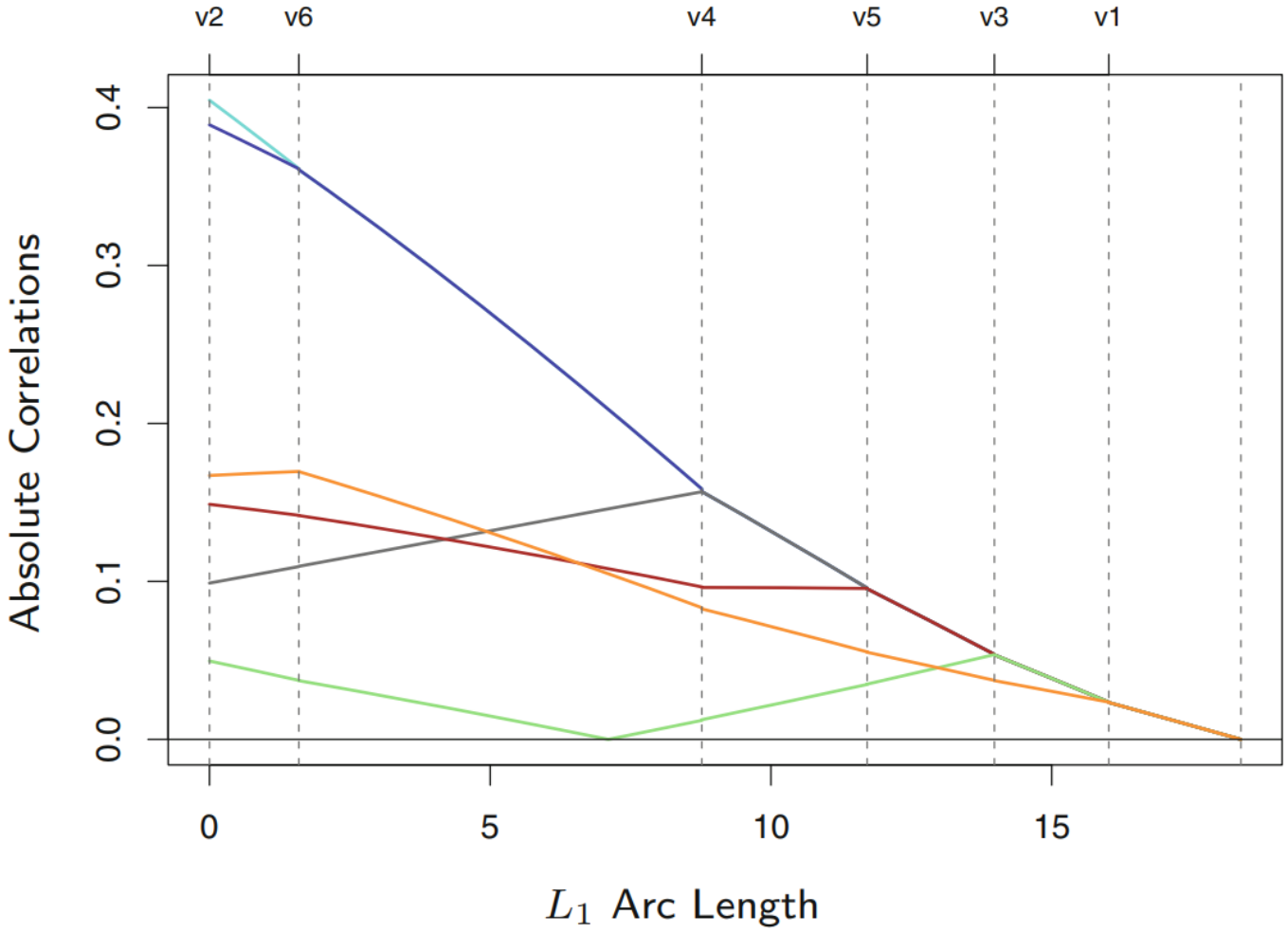
- Suppose $\mathcal{A}_k$ is the active set of variables at the beginning of the kth step, and let $\beta_{\mathcal{A}_k}$ be the coefficient vector for these variables at this step; there will be k - 1 nonzero values, and the one just entered will be zero. If $\mathbf{r}_k = \mathbf{y} - \mathbf{X}_{\mathcal{A}_k}\beta_{\mathcal{A}_k}$ is the current residual, then the direction for this step is

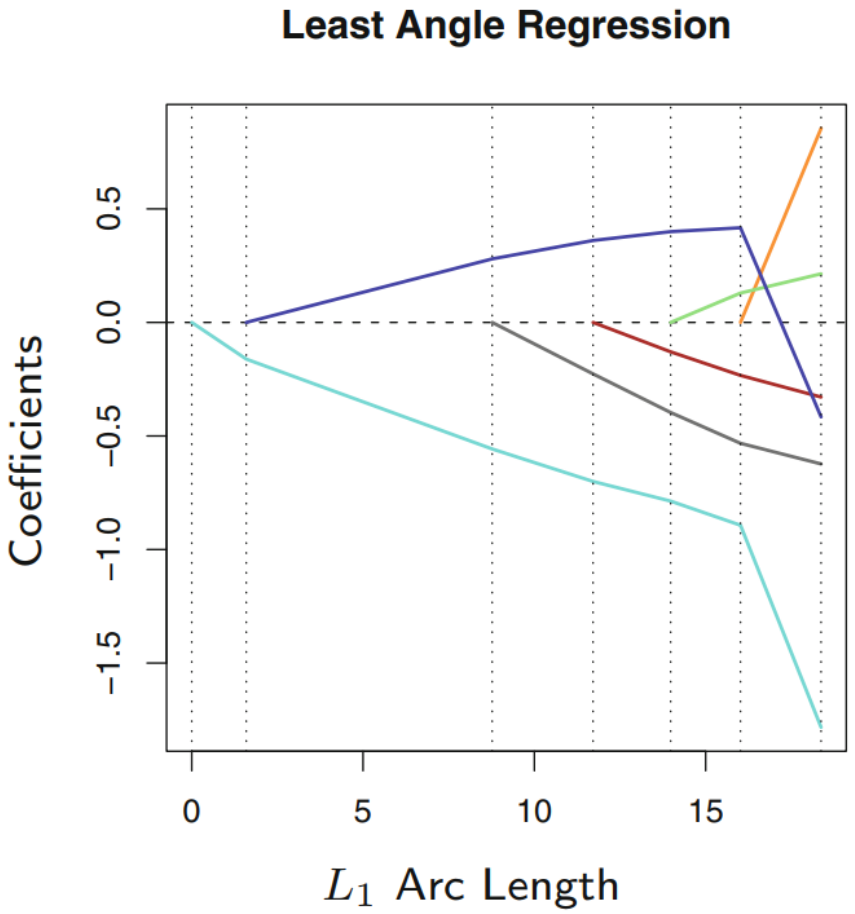
$$\delta_k = \left(\mathbf{X}_{\mathcal{A}_k}^T \mathbf{X}_{\mathcal{A}_k}\right)^{-1} \mathbf{X}_{\mathcal{A}_k}^T \mathbf{r}_k$$

- The coefficient profile then evolves as $\beta_{\mathcal{A}_k}(\alpha) = \beta_{\mathcal{A}_k} + \alpha \cdot \delta_k$.
- If the fit vector at the beginning of this step is $\hat{\mathbf{f}}_k$, then it evolves as $\hat{\mathbf{f}}_k(\alpha) = \hat{\mathbf{f}}_k + \alpha \cdot \mathbf{u}_k$, where $\mathbf{u}_k = \mathbf{X}_{\mathcal{A}_k}\delta_k$.

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LARS solution path





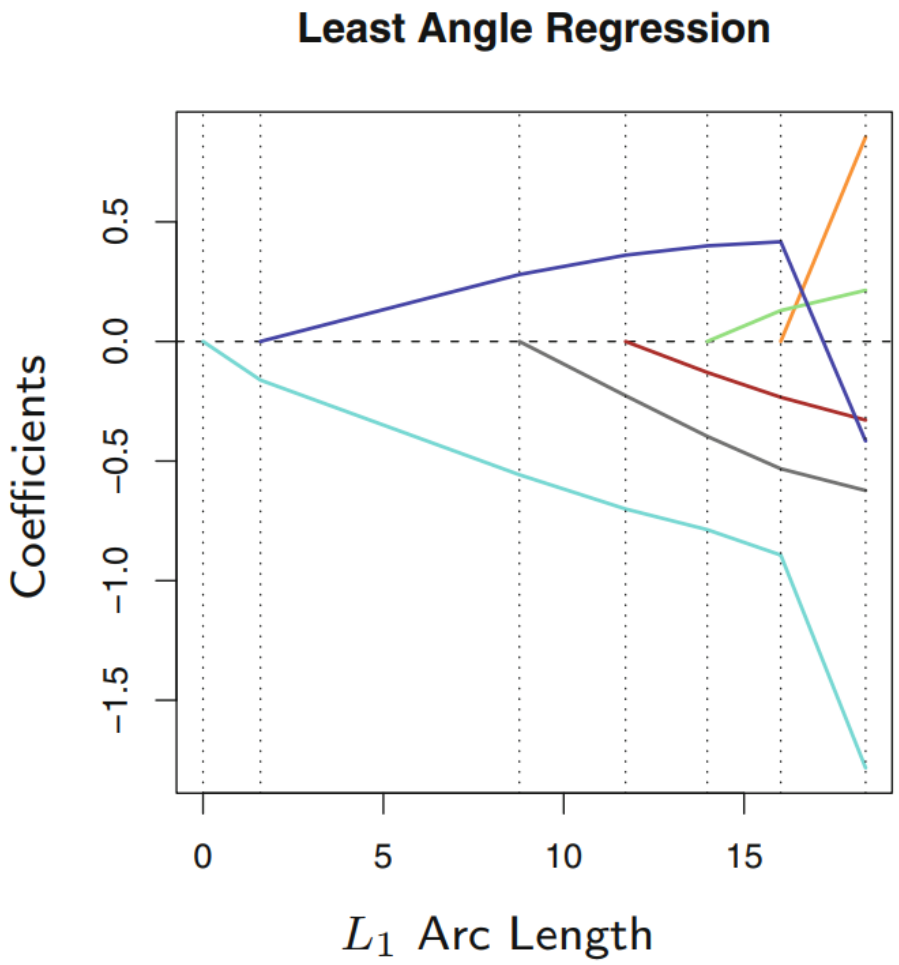
Computation

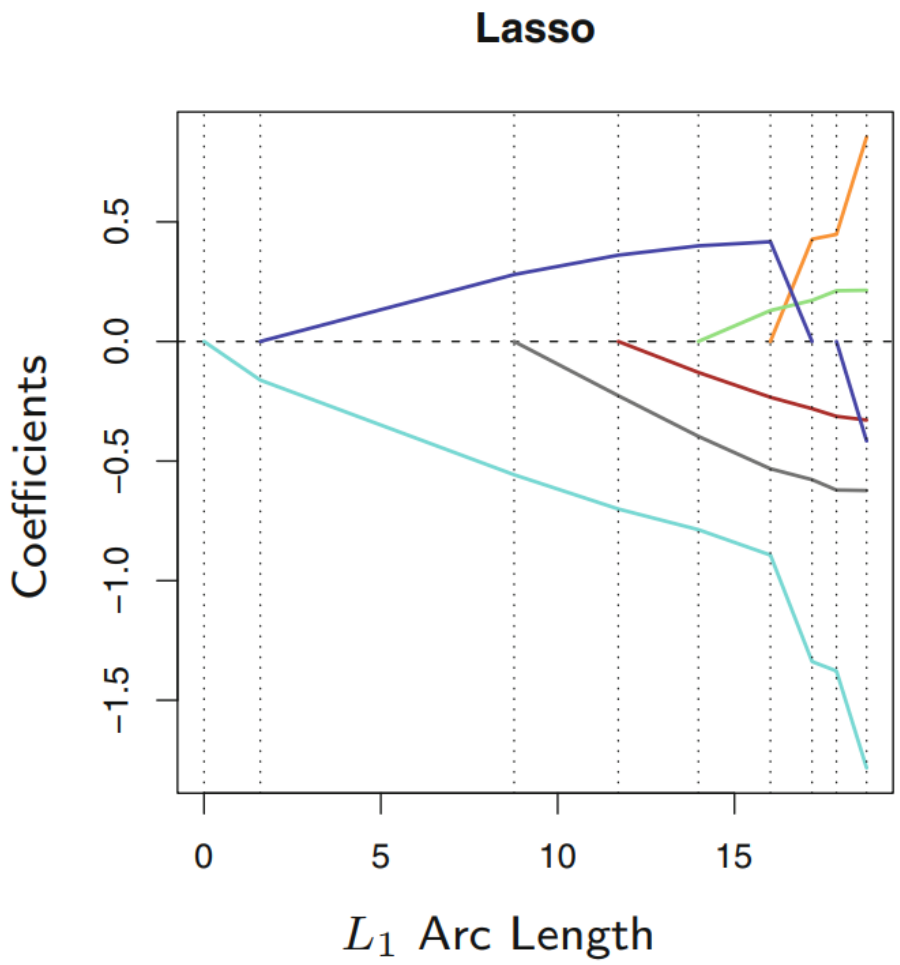
- LARS for Lasso

4. … Additionally, if a non-zero coefficient hits zero, drop its variable from the active set of variables and recompute the current joint least squares direction.

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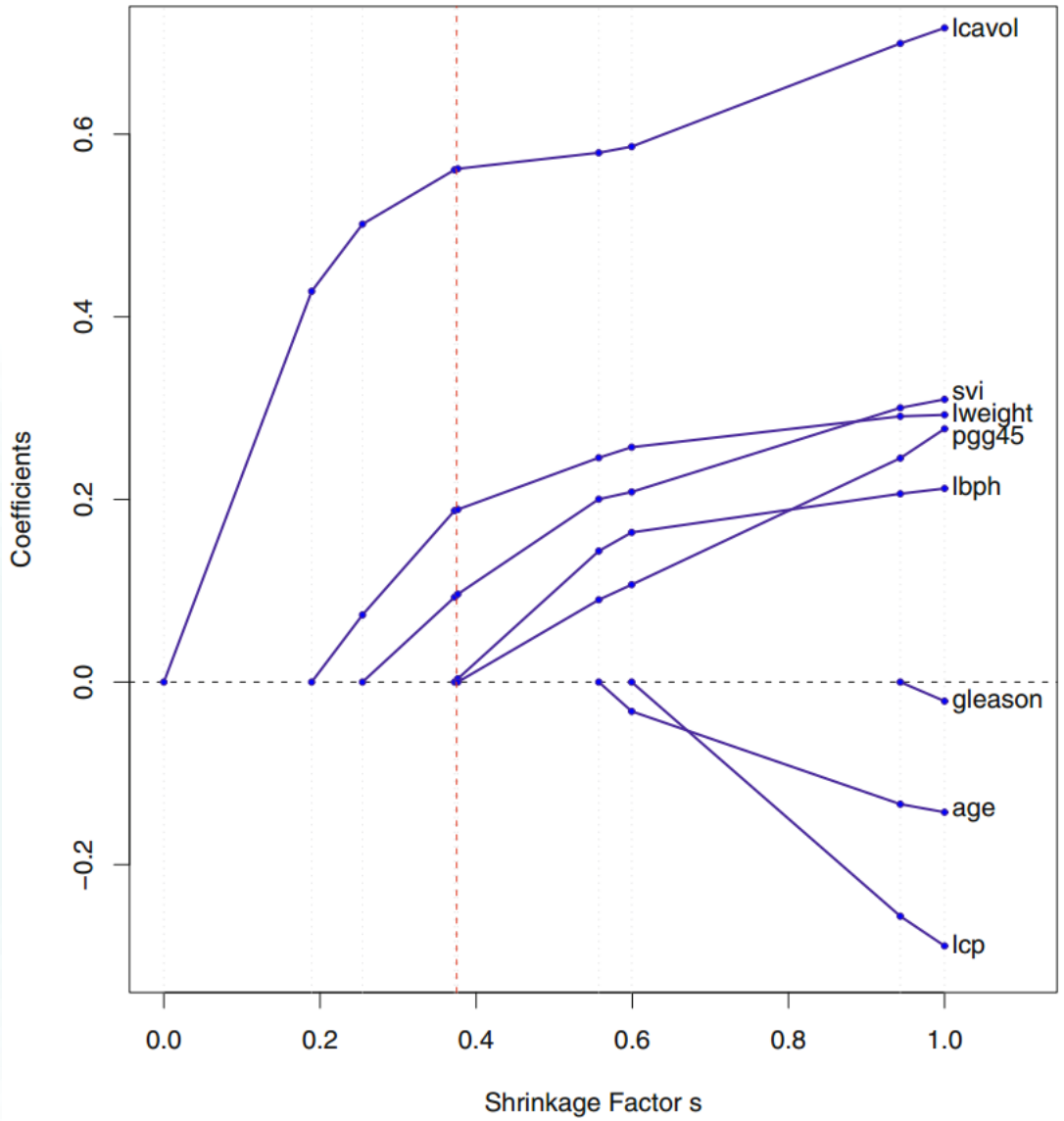
Lasso solution path



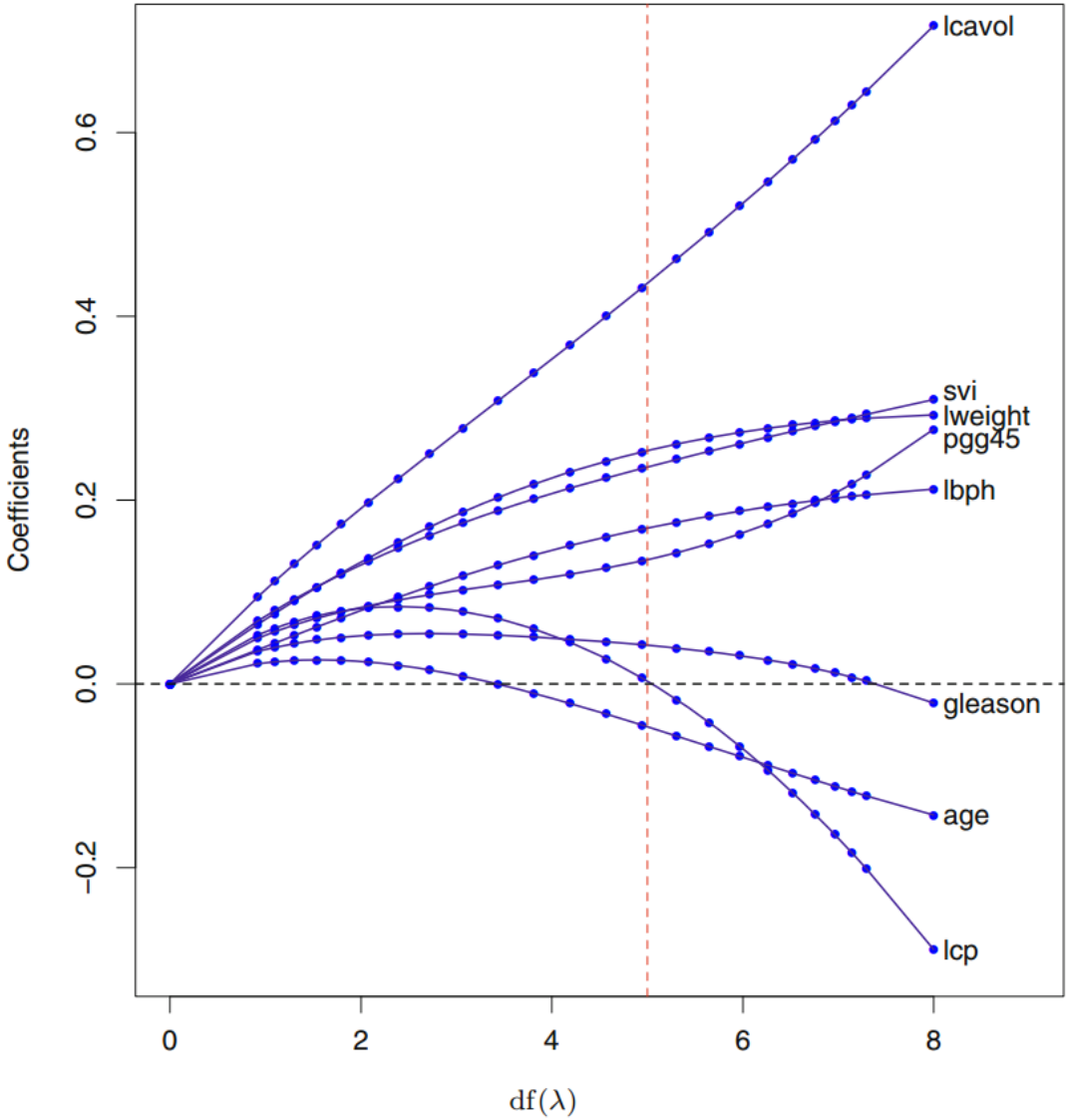


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Lasso Solution Path



Lasso Solution Path



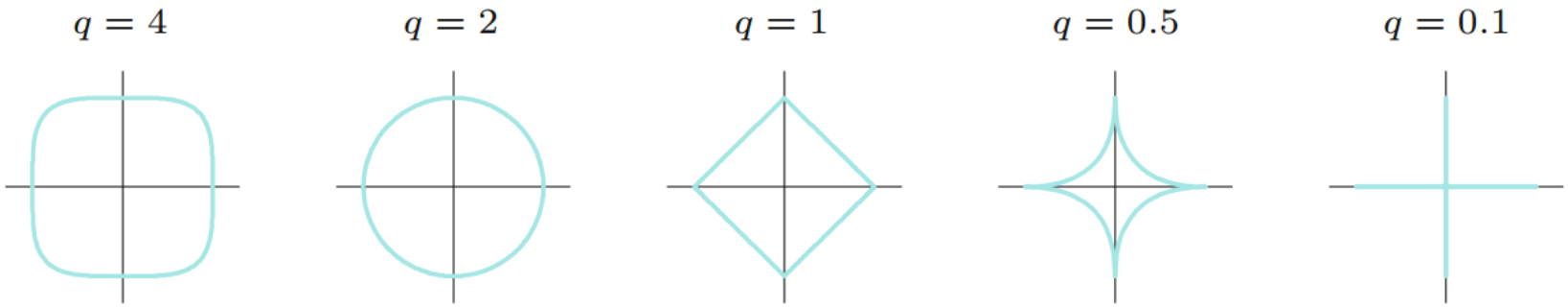
Other Regularized Models

Generalized L_q Penalty

$$\min_{\beta} \sum_{i=1}^{N} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$
$$\text{s.t. } \sum_{j=1}^{p} |\beta_j|^q \leq s$$

- $q = 0$ corresponds to best subset selection.
- For $q \leq 1$, more weights are imposed on the coordinate directions.
- $q = 1$ is the smallest q such that the constraint region is convex.

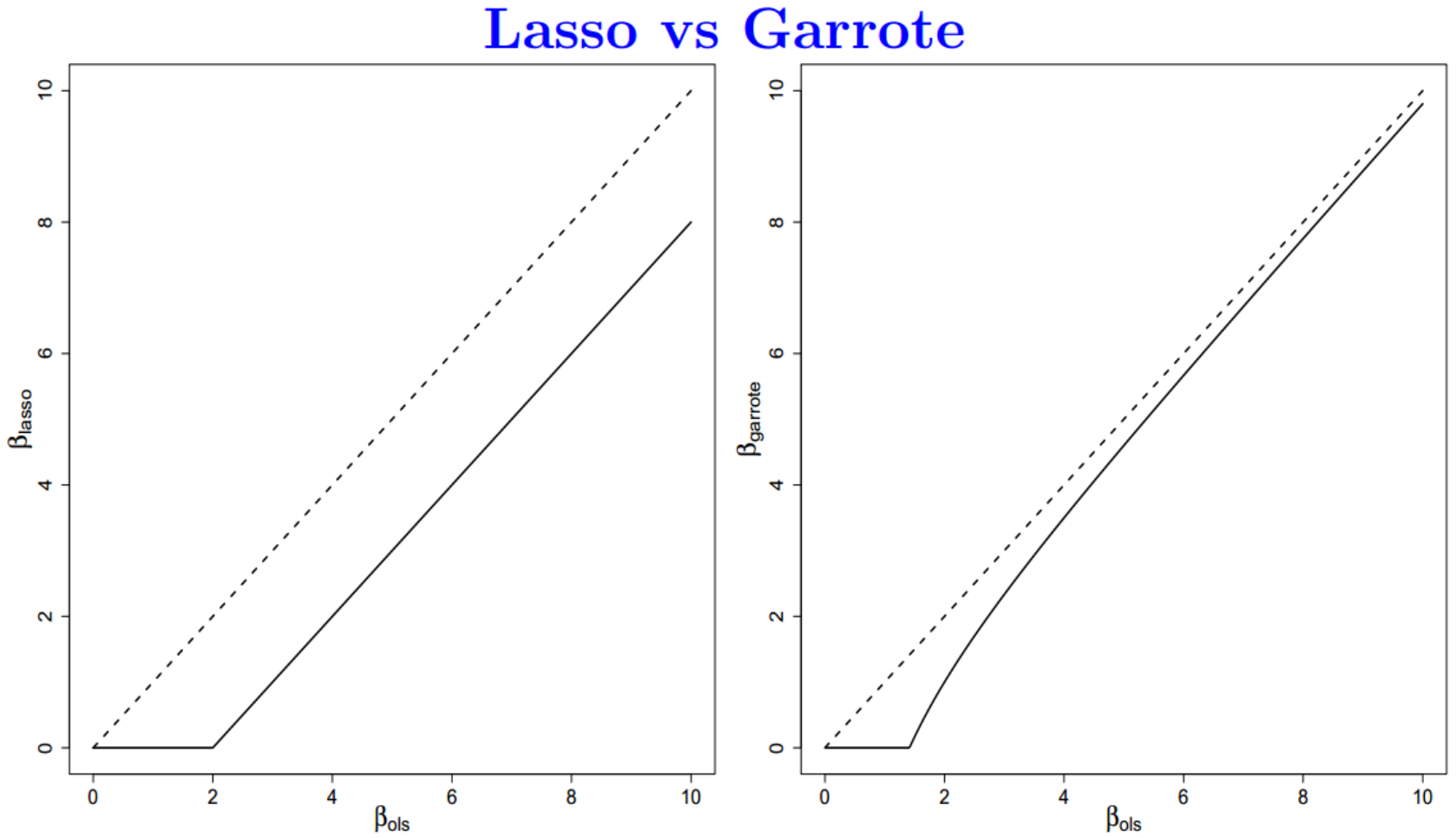
Generalized L_q Penalty



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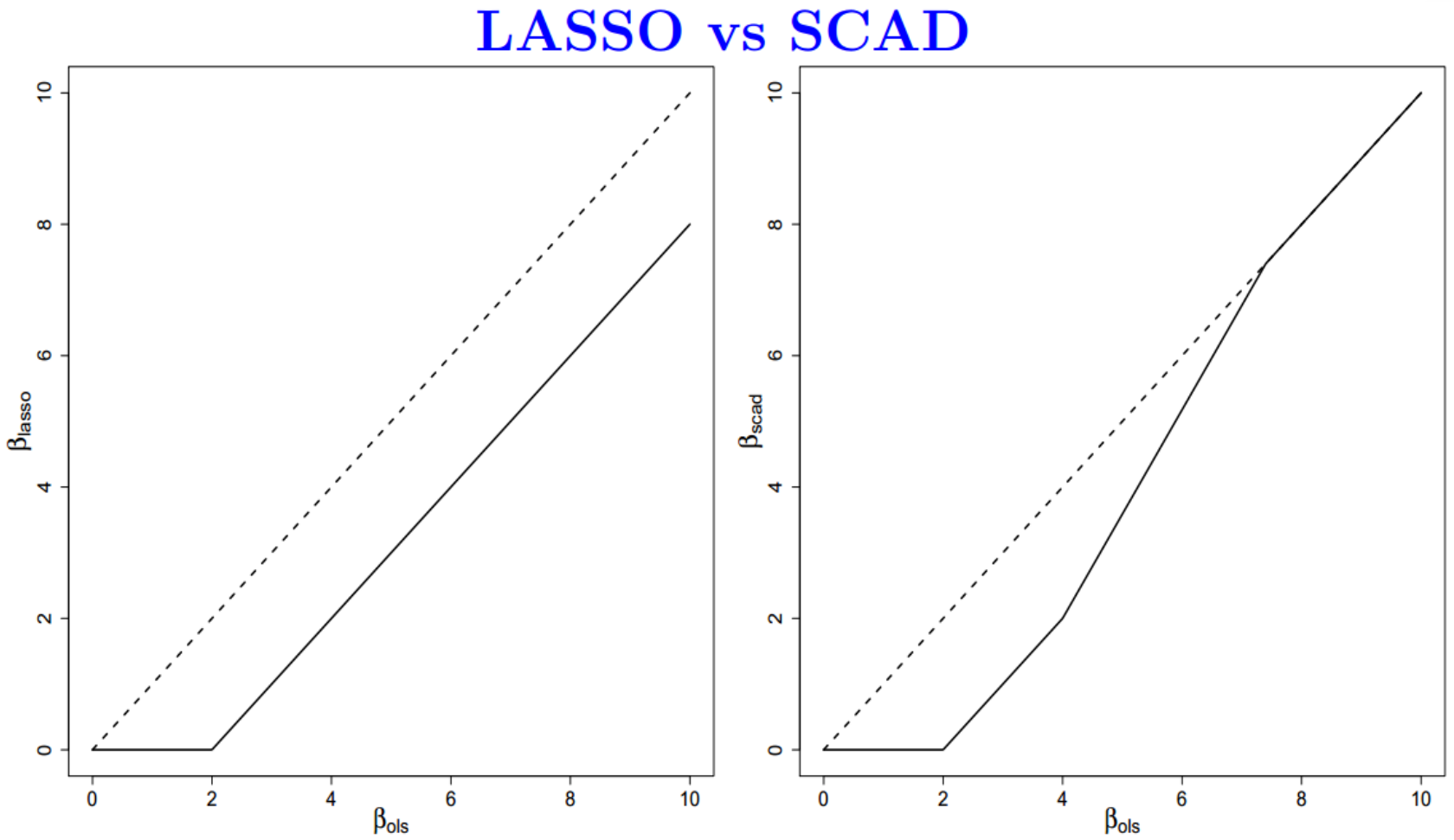
Non-negative Garrote

- Breiman (1994)



SCAD

- Fan and Li (2001)



Elastic-Net

- Zou and Hastie (2005)

$$\hat{\beta}^{enet} = \underset{\beta}{\operatorname{argmin}}(y - X\beta)^T(y - X\beta) + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2^2.$$

or equivalently,

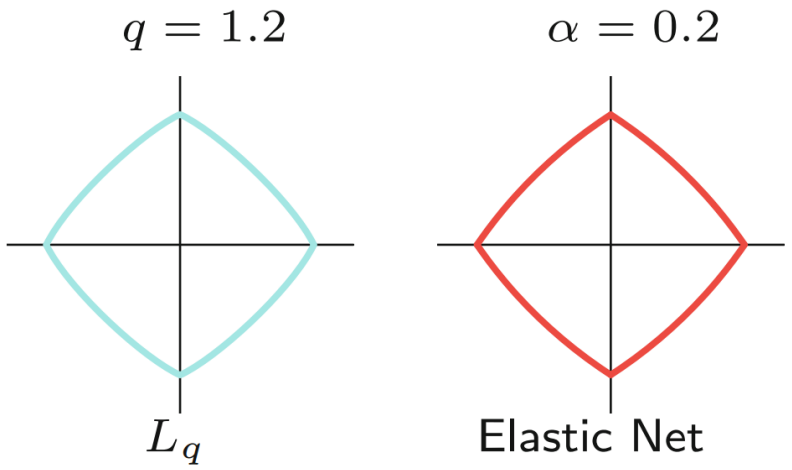
$$\hat{\beta}^{enet} = \underset{\beta}{\operatorname{argmin}}(y - X\beta)^T(y - X\beta)$$

$$\text{s.t. } (1-\alpha)\|\beta\|_1 + \alpha\|\beta\|_2^2 \leq s,$$

where $\alpha = \lambda_2/(\lambda_1 + \lambda_2)$.

Elastic-Net

- L1 penalty encourages sparsity.
- Ridge penalty
  - Able to select more than *n* variables.
  - Encourages “grouping effect”—strongly correlated predictors tend to be in or out of the model together.



Oracle Properties

- As $n \to \infty$, the regression procedure
  - selects the right model with probability tending to one.
  - coefficient estimate has the same asymptotic distribution as if the true model was known in advance.

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Oracle Properties

Regression Technique	Oracle Property
LASSO	NO
Elastic-Net	NO
Garrote	YES
SCAD	YES

Adaptive LASSO

- Zou (2006)

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^{p} w_j |\beta_j| ,$$

where w_j are the adaptive data-driven weights and can be computed by

$$w_j = \left| \hat{\beta}_j^{ini} \right|^{-\gamma} .$$

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Adaptive LASSO

- In practice, usually set

$$w_j = \left|\hat{\beta}_j^{ols}\right|^{-1},$$

i.e.

$$\hat{\beta} = \underset{\beta}{\operatorname{argmin}} \sum_{i=1}^{n} \left(y_i - \sum_{j=1}^{p} \beta_j x_{ij}\right)^2 + \lambda \sum_{j=1}^{p} \frac{|\beta_j|}{\left|\hat{\beta}_j^{ols}\right|}.$$

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Adaptive Elastic-Net

- Zou and Zhang (2009)

$$\min_{\beta} \|y - X\beta\|_2^2 + \lambda_2 \|\beta\|_2^2 + \lambda_1 \sum_{i=1}^{p} \frac{|\beta_j|}{|\hat{\beta}_j^{enet}|}$$

Oracle Properties

Regression Technique	Oracle Property
LASSO	NO
Elastic-Net	NO
Garrote	YES
SCAD	YES
Adaptive LASSO	YES
Adaptive Elastic-Net	YES

Group Lasso

- Yuan and Lin (2006)

$$\min_{\beta}\left\{\left\|y-\sum_{j=1}^{J}X_j\beta_j\right\|_2^2+\lambda\sum_{j=1}^{J}\left(\beta_j^T K_j\beta_j\right)^{1/2}\right\}$$

- X and β are divided into groups. $K_j, j=1,\ldots,J$, is positive definite.
- Feature selection by the groups.
- When $J=p$ and $K_j=I$, group lasso becomes standard lasso.
- When $J=1$ and $K_1=I$, group lasso becomes ridge regression.

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Fused Lasso

- Tibshirani et al. (2005)

$$\min_{\beta} \quad \|y - X\beta\|_2^2$$

$$\text{s.t.} \quad \sum_{j=1}^{p} |\beta_j| \le s_1$$

$$\sum_{j=2}^{p} |\beta_j - \beta_{j-1}| \le s_2$$